（装订线内不要答题）

复旦大学大数据学院

2022 ~ 2023 学年第一学期期末考试试卷

☒ A 卷　　☐ B 卷

课程名称：**数据结构**
课程代码：**DATA130023.01**
开课院系：**大数据学院 考试形式**：**闭卷**
考试时间：**90 分钟**
姓　名：_____ **学　号**：_____ **专　业**：_____

声明：我已知悉学校对于考试纪律的严肃规定，将秉持诚实守信宗旨，严守考试纪律，不作弊，不剽窃；若有违反学校考试纪律的行为，自愿接受学校严肃处理。

学生（签名）：_____
年　　月　　日

题　目	一	二	三	四	五	六	总　分
得　分							

- When writing an algorithm, a clear description will suffice. Pseudo-code is not required.

一、**(24pts) True** or **False**?

(i) **(4pts)** If the graph is a DAG, then a DFS traversal of the graph will not find any back edges. [　　]

(ii) **(4pts)** If we modify the Kosaraju algorithm to replace the second depth-first search with breadth-first search, then it will still find the strong components. [　　]

(iii) **(4pts)** If a graph has negative-weight edges, there exists a constant C such that adding C to every edge weight and running Dijkstra's algorithm produces shortest paths under the original edge weights　　[　　]

(iv) **(4pts)** Let T be a minimum spanning tree of G. Then, for any pair of vertices s and t, the shortest path of s to t in G is the path from s to t in T.　　[　　]

(v) **(4pts)** Topological sort can be solved by DFS.　　[　　]

(vi) **(4pts)** There is a polynomial-time algorithm for the Knapsack problem if all items have size in $\{0, 1\}$.　　[　　]

二、**(16pts)** Algorithm Design Techniques.

(i) **(8pts)** The following are a few of the design strategies we followed in class to solve several problems: **1. Dynamic programming. 2. Greedy. 3. Divide-and-conquer.** For each of the following problems, mention which of the above design strategies was used (in class) in the following algorithms.

(1) Huffman coding. _____
(2) Longest common subsequences algorithm. _____
(3) Median Selection. _____
(4) Knapsack problem. _____

(ii) **(8pts)** The following are a few of the data structures we discussed: **1. Queue. 2. Stack. 3. Priority Queue**. For each of the following problems, mention which of the above data structures was used (in class) in the following algorithms.

(1) Dijkstra's algorithm for shortest path. _____
(2) Breadth first search. _____
(3) Depth first search. _____
(4) Prim's algorithm. _____

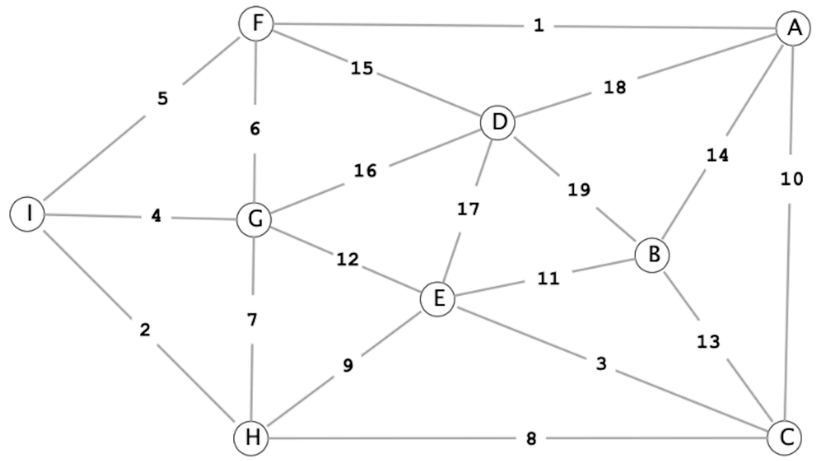
三、**(15pts)**

(i) **(8pts)** What is an optimal Huffman code for the following set of frequencies, based on the first 8 Fibonacci numbers?
a:1 b:1 c:2 d:3 e:5 f:8 g:13 h:21

(ii) **(7pts)** Generalize your answer to find the optimal code when the frequencies are the first n Fibonacci numbers.

（装订线内不要答题）

四、**(15pts)** Consider the following weighted graph with 9 vertices and 19 edges.



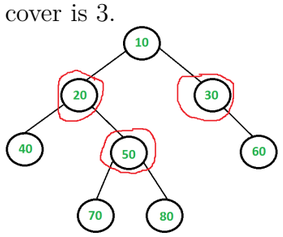
(i) **(7.5pts)** Present the sequence of edges in the MST in the order of Kruskal's algorithm (by specifying their edge weights).

(ii) **(7.5pts)** Present the sequence of edges in the MST in the order of Prim's algorithm (by specifying their edge weights). Assume the first edge added has weight 1.

五、**(15pts)** Prove that a directed acyclic graph must have at least one vertex that has no incoming edges.

（装订线内不要答题）

六、**(15pts)** A vertex cover of an undirected graph is a subset of its vertices such that for every edge (u, v) of the graph, either u or v (or both) is in vertex cover.
In this problem, the input graph is always a Binary Tree; your task is to design an algorithm to find a vertex cover with smallest number of vertices. For example, consider the following binary tree. The smallest vertex cover is $\{20, 50, 30\}$ and size of the vertex cover is 3.



i **(8pts)** For a vertex u, let $c(u)$ denote the size of the smallest vertex cover of the subtree rooted at u. Now suppose $c(v)$ is known for every descendant of u, give a recurrence formula for $c(u)$.

ii **(7pts)** Based on the recurrence above, describe a dynamic programming algorithm which outputs the smallest vertex cover size of the input tree.